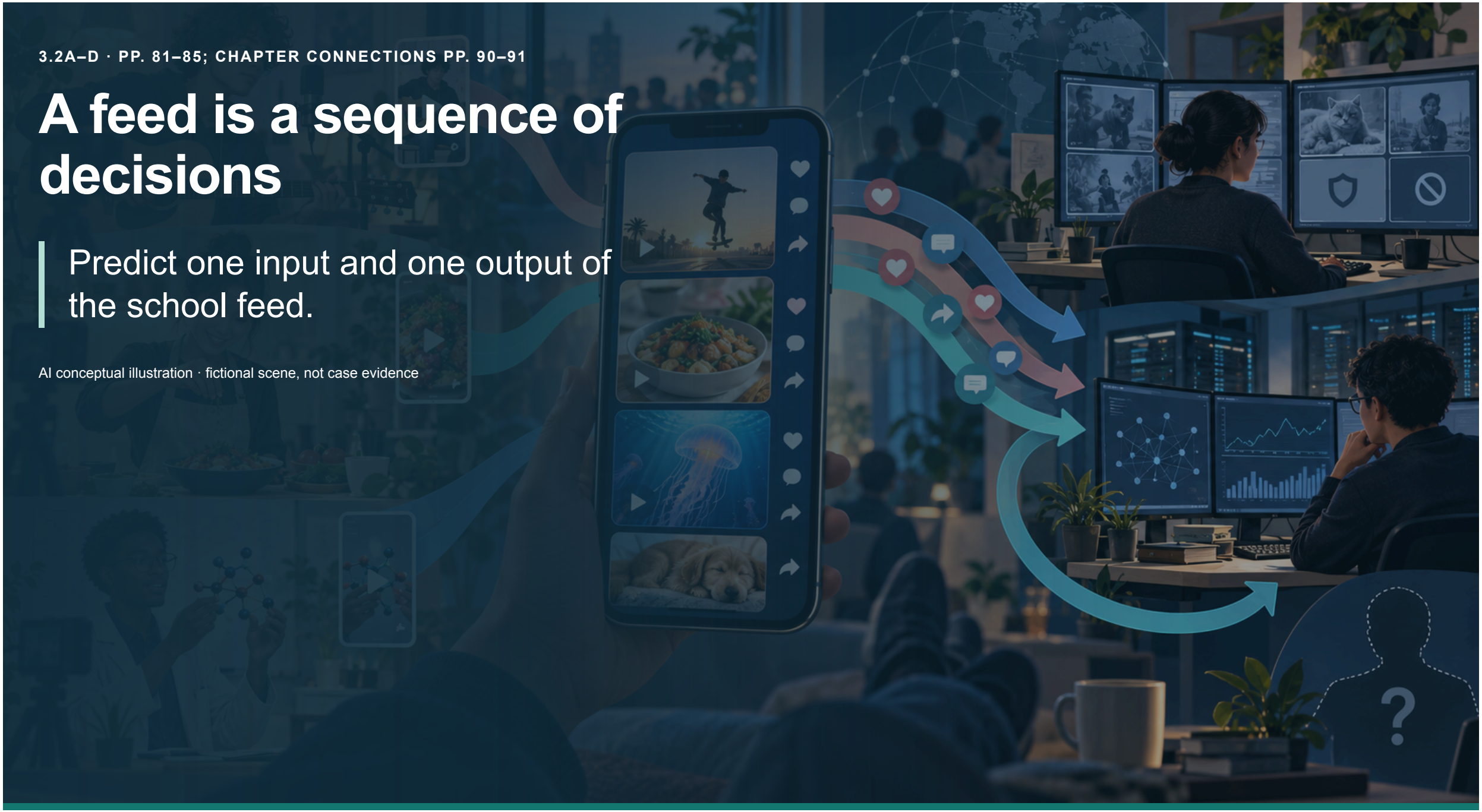
3.2A–D · PP. 81–85; CHAPTER CONNECTIONS PP. 90–91

A feed is a sequence of decisions

> Predict one input and one output of the school feed.



AI conceptual illustration · fictional scene, not case evidence

An algorithm

01

Input: B4, B1, B3, B2

→

02

Compare each record with target B3

→

03

First match → return B3 and stop

A concrete procedure has defined information, a precise operation and a stopping condition.

A precise sequence of instructions for a task

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81–82.

Characteristics of a usable algorithm

AMBIGUOUS INSTRUCTION

Show the best item.

FOLLOWABLE RULE

Sort by completion; break ties by item ID; stop at list end.

Unambiguous · finite · feasible · independent of a particular coding language

Unambiguous: each instruction has one clear meaning

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81–82.

APPLY · 4 · TEXTBOOK PP. 82

Variables hold values

Step	Instruction	total
Start	Set total to 0	0
Item A	Add 1	1
Item B	Add 1	2

A variable is a named place for a value

Conditionals choose a branch

duration = 60 seconds

duration ≤ 60?

TRUE → include

60 meets the boundary

FALSE → exclude

75 fails the test

Missing value → separate rule; missing ≠ 0

IF a test is true, follow one branch

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 83.

Loops repeat a set of instructions

WORKED TRACE · SYNTHETIC DURATIONS

30 s 75 s 60 s

Start count = 0 · Test duration ≤ 60 · Advance even when false

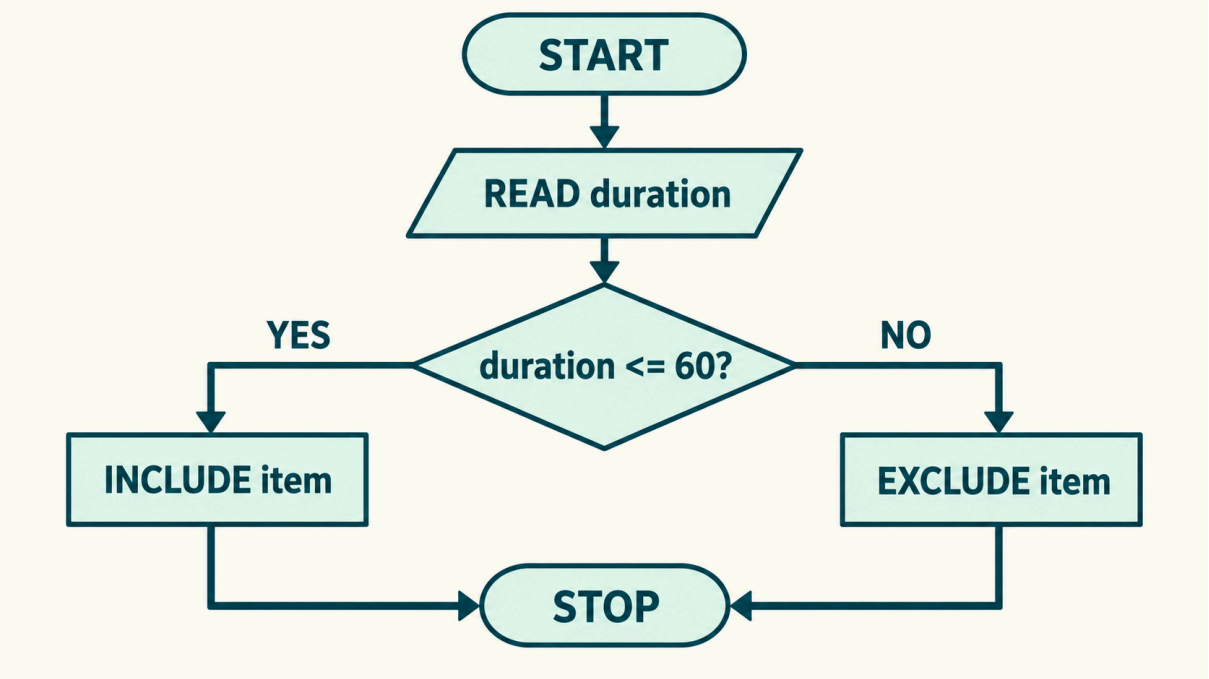
30 → count 1; 75 → count 1; 60 → count 2; end → stop.

Watch the counter and current item. Explain why the second item does not add one.

FOR each item, apply the same test

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 83.

One procedure, several representations



Original teaching diagram

Natural language explains a procedure in words

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 83.

Algorithm, pseudocode and code

PROCEDURE

If a video is at most 60 seconds, say Include; otherwise say Exclude.

PSEUDOCODE

IF seconds ≤ 60
THEN say Include
ELSE say Exclude

PYTHON CODE

```
if seconds <= 60:
    print("Include")
else:
    print("Exclude")
```

The code fragment assumes a numeric seconds value. The same decision is expressed in words, a planning notation and a specific language.

Algorithm: the underlying procedure

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 83–84.

APPLY · 9 · TEXTBOOK PP. 84

Searching finds a target

Comparison	Record	Match B3?
1	B4	No
2	B1	No
3	B3	Yes: stop

Linear search checks records one at a time

Sorting changes the order

State	Values	Action
Start	3, 1, 2	Compare first pair
After swap	1, 3, 2	Compare next pair
After swap	1, 2, 3	End this pass

Sort arranges existing items by a stated key

Filtering and counting

Task	Output
Filter loans >= 10	A and C
Count matches	2 records
Sort all, descending loans	C, A, B

Filter: keep records that meet a condition

Efficiency needs a measure

RESOURCE MEASURE

3 comparisons

B3 in B4, B1, B3, B2

INPUT CONDITION

Sorted or unsorted?

State the assumptions

SOCIAL CRITERION

Fair access?

Speed alone cannot establish this

Name the measure before saying an algorithm is better.

Possible measures: comparisons, time or memory

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81, 84–86.

Prioritizing turns a goal into an order

Item	Completion c	New topic m
M1	0.8	1
M2	0.9	0
M3	0.5	0

A ranking score combines chosen signals

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 85.

APPLY · 14 · TEXTBOOK PP. 85

A changed objective: prediction

SAME INPUTS

M1: c=.8, m=1

M2: c=.9, m=0

M3: c=.5, m=0

RULE A · DISCOVERY

c + 2m

The new-topic flag contributes up to 2.

RULE B · COMPLETION

3c + .2m

Completion contributes up to 3.

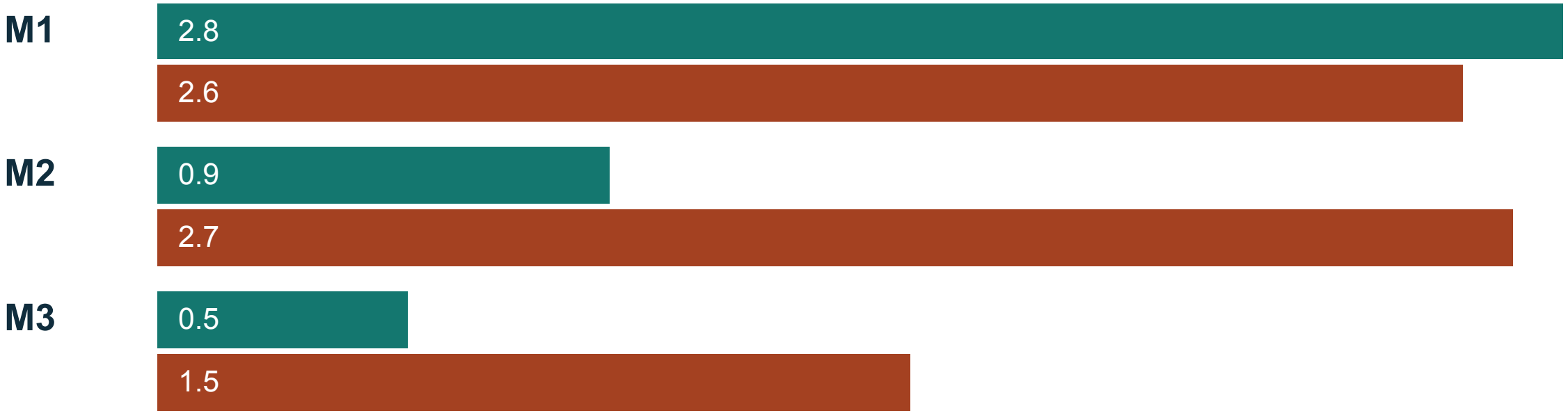
Predict first. Calculate all three scores. The next slide reveals the ranking.

Rule A: score = c + 2m

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 85.

A changed objective: worked result

Synthetic score · teal: Rule A · rust: Rule B · common scale 0–2.8



Rule A ranks M1 first

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 85.

APPLY · 16 · TEXTBOOK PP. 85

Classifying and associating

CLASSIFICATION

Ticket → high / medium / low urgency

ASSOCIATION

Bread and soup → often in the same basket

A label assigns a category. Co-occurrence identifies a pattern, not a cause.

Classifying assigns records to a category

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 85.

Rules and learned models

SPECIFIED RULE

Designer chooses formula and weights

LEARNED MODEL

Examples fit internal parameters to a chosen objective

Human choices remain: training data, objective and evaluation.

A hand-written rule specifies the calculation

The complete decision pipeline

1. Collect/validate →

2. Filter eligible items →

3. Calculate score →

4. Rank and display

A response after display can become a later input. The rule and objective must be stated, not guessed.

Collect and validate the input

Independent transfer: a library queue

DEMAND

More requests than copies

Fictional library

CHOOSE AN OBJECTIVE

First request? Need? Waiting time?

Defend the priority

TRACE THE RULE

Input → condition → stop → output

Name who could lose priority

Independent transfer: close notes; produce four sentences with a trade-off.

A school library prioritizes reservation requests

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81–85.

APPLY · 20 · TEXTBOOK PP. 81–91

Chapter 3.2: the knowledge to carry forward

PAUSE · INDIVIDUAL FIRST ATTEMPT · 2 MINUTES

Which explanation demonstrates a mechanism?

Predict / retrieve | Justify with an operation | Check → correct

Save your first response below. Explain the correction on a fresh example; practice is not a mark.

Describe characteristics and components

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81–91.